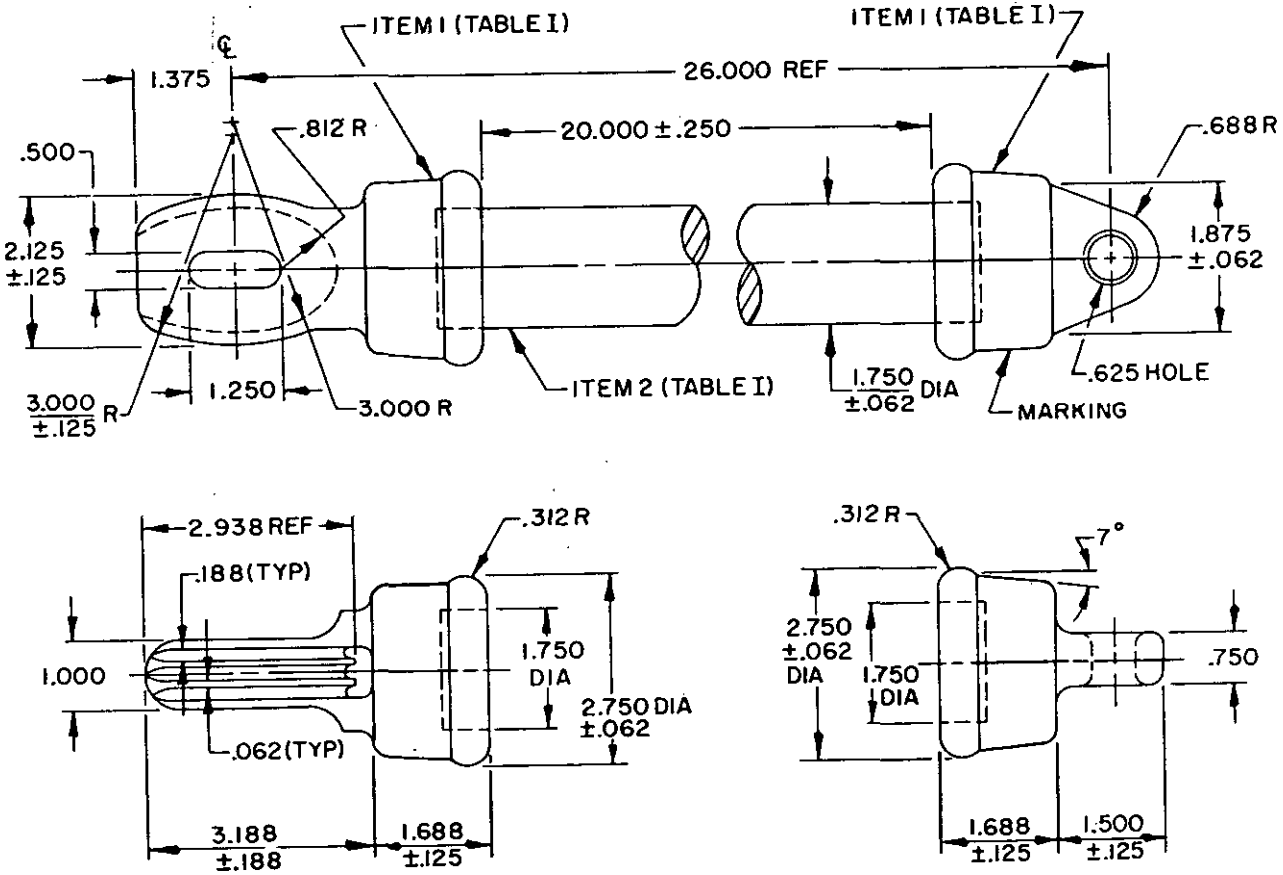
MIL-I-23264/21B
25 August 1977
SUPERSEDING
MIL-I-23264/21A
8 July 1968

MILITARY SPECIFICATION SHEET

INSULATORS, STRAIN
(STYLES 68, 69, 70, AND 71)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the insulators described herein shall consist of this document and the latest issue of Specification MIL-I-23264.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.062	1.57	.625	15.88	1.000	25.40	1.875	47.62	20.000	508.00
.125	3.18	.688	17.48	1.250	31.75	2.125	53.98	26.000	660.40
.188	4.78	.750	19.05	1.375	34.92	2.750	69.85		
.312	7.92	.812	20.62	1.688	42.88	2.938	74.63		
.500	12.70	.875	22.22	1.750	44.45	3.188	80.98		

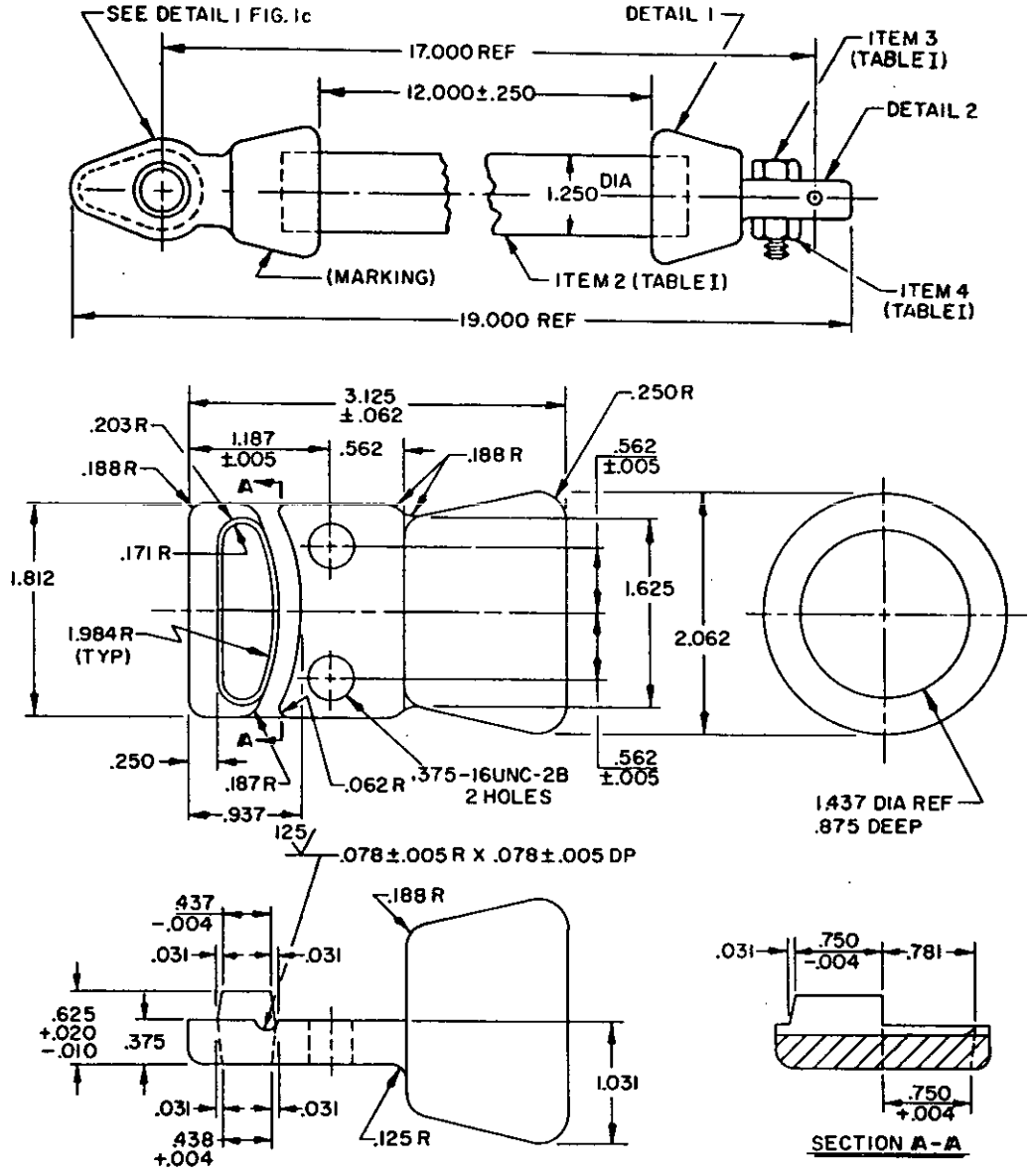
FIGURE 1a. Strain insulator (NL422B68-208) (NL422W68-208).

(B) FIGURE 1. Styles 68, 69, 70 and 71 strain insulators.

(B) denotes changes

FSC 5970

Source: https://assist.dla.mil -- Downloaded: 20261O-19T16:09Z
Check the source to verify that this is the current version before use.

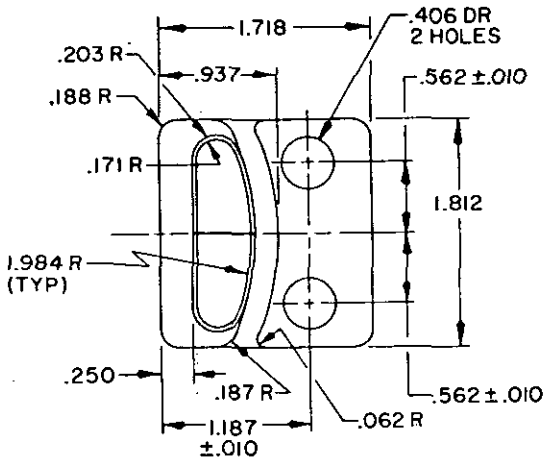


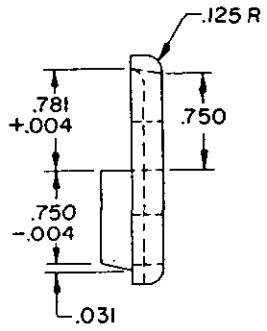
DETAIL I
(Item 1, Table I)

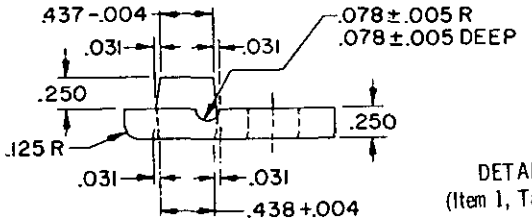
INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.004	.10	.078	1.98	.250	6.35	.750	19.05	1.437	36.50
.005	.13	.125	3.18	.375	9.52	.781	19.84	1.625	41.28
.010	.25	.171	4.34	.437	11.10	.875	22.22	1.812	46.02
.020	.51	.187	4.75	.438	11.13	1.031	26.19	1.984	50.39
.031	.79	.188	4.78	.562	14.27	1.187	30.15	2.062	52.37
.062	1.57	.203	5.16	.625	15.88	1.250	31.75	3.125	79.38
								12.000	304.80
								17.000	431.80

(B) FIGURE 1. Styles 68, 69, 70 and 71, strain insulators. - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:09Z
Check the source to verify that this is the current version before use.







DETAIL 2
(Item 1, Table 1)

INCHES	MM	INCHES	MM
.004	.10	.406	10.31
.005	.13	.437	11.10
.010	.25	.438	11.13
.031	.79	.562	14.27
.062	1.57	.750	19.05
.078	1.98	.781	19.84
.125	3.18	.937	23.80
.171	4.34	1.187	30.15
.187	4.75	1.718	43.64
.188	4.78	1.812	46.02
.203	5.16	1.984	50.39
.250	6.35		

FIGURE 1b. Strain insulator (NL422B69–136) (NL422W69–136) - Continued.

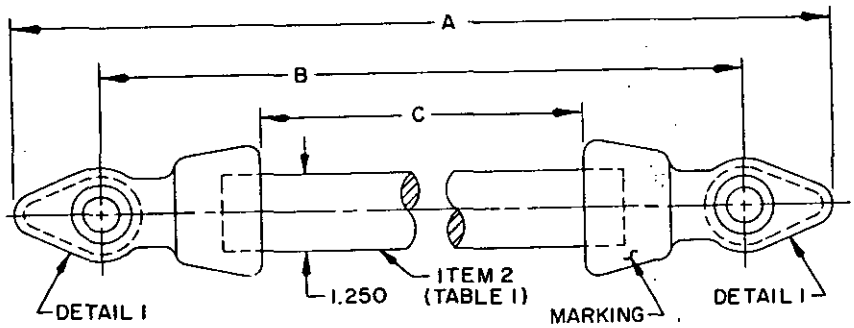
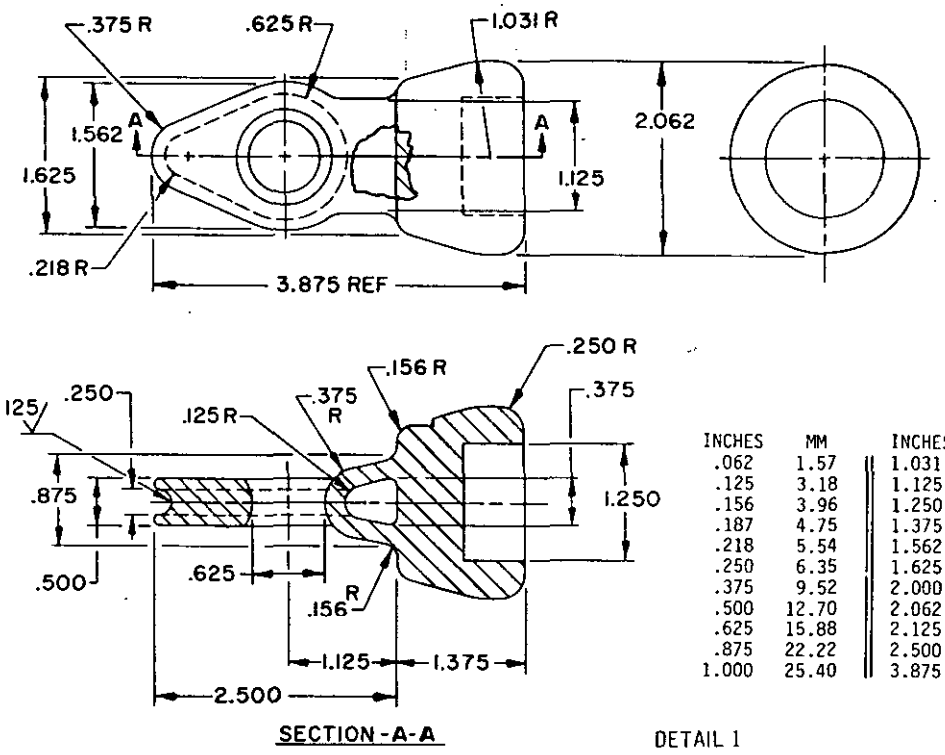


FIGURE 1c. Strain insulator (NL422B70–104 NL422B70–168) (NL422W70–104 NL422W70–168)

(B) FIGURE 1. Styles 68, 69, 70 and 71, strain insulators. - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:09Z
Check the source to verify that this is the current version before use.



INCHES	MM	INCHES	MM
.062	1.57	1.031	26.19
.125	3.18	1.125	28.58
.156	3.96	1.250	31.75
.187	4.75	1.375	34.92
.218	5.54	1.562	39.67
.250	6.35	1.625	41.28
.375	9.52	2.000	50.80
.500	12.70	2.062	52.37
.625	15.88	2.125	53.98
.875	22.22	2.500	63.50
1.000	25.40	3.875	98.42

DETAIL 1
(Item 1, Table I)

FIGURE 1c. Strain insulator (NL422B70-104 NL422B70-168) (NL422W70-104 NL422W70-168) - Continued.

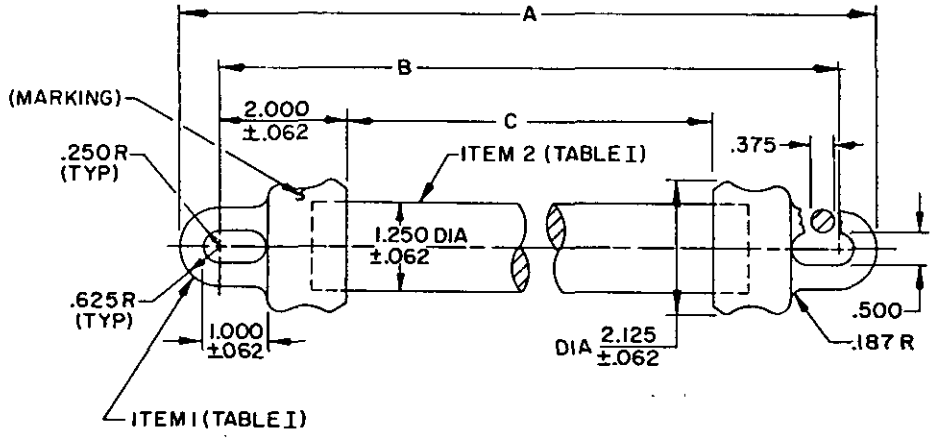
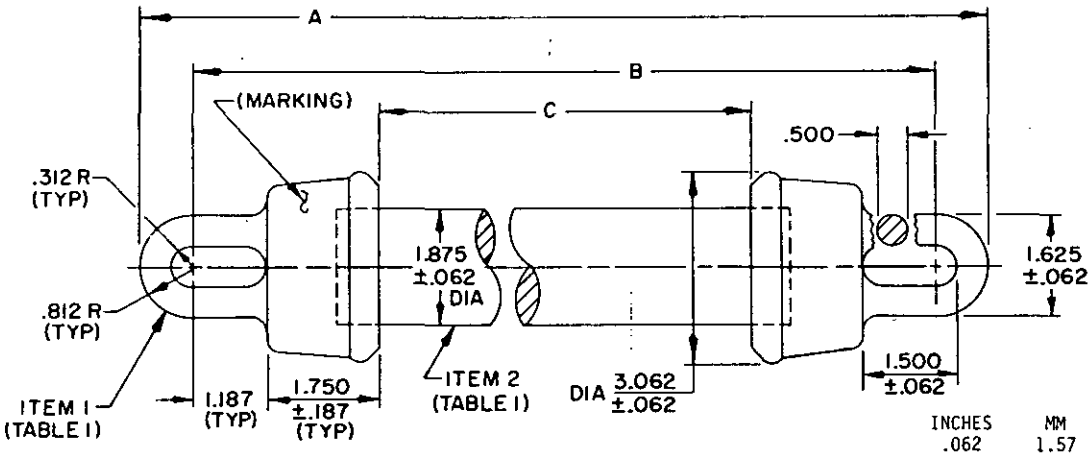


FIGURE 1d. Strain insulator (NL422B71-072 NL422B71-112) (NL422W71-072 NL422W71-112)

(B) FIGURE 1. Styles 68, 69, 70 and 71, strain insulators. - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:09Z
Check the source to verify that this is the current version before use.



INCHES	MM
.062	1.57
.250	6.35
.312	7.92
.500	12.70
.812	20.62
.875	22.22
1.187	30.15
1.500	38.10
1.625	41.28
1.750	44.45
1.875	47.62
3.062	77.77

FIGURE 1e. Strain insulator (NL422B71-167 NL422B71-207) (NL422W71-167 NL422W71-207)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

(B) FIGURE 1. Styles 68, 69, 70 and 71, strain insulators. - Continued.

TABLE I. Item description.

Item No.	Description	Material	No. per assembly
1	Cap	Brass, composition 1, hard, per QQ-B-637, or brass, class A, per QQ-B-621; or bronze, composition 1, per QQ-L-225	1
2	Insulator	Ceramic, grade 422 or better, per MIL-I-10	1
3	Bolt, 3/8-16UNC-2A, 1-1/4 in. long	Brass	2
4	Hexagon nut, 3/8-16UNC-2B	Brass	2

Source: https://assist.dla.mil -- Downloaded: 202610-19T16:09Z
Check the source to verify that this is the current version before use.

(B) TABLE II. Type designation.

Type designation	Dimensions 1/		
	A	B	C
NL422B68-208 NL422W68-208	See figure 1a		
NL422B69-136 NL422W69-136	See figure 1b		
NL422B70-104 NL422W70-104	15.750 (400.05)	13.000 (330.20)	8.000 (203.20)
NL422B70-168 NL422W70-168	23.750 (603.25)	21.000 (533.40)	16.000 (406.40)
NL422B71-072 NL422W71-072	10.250 (260.35)	9.000 (228.60)	5.000 (127.00)
NL422B71-112 NL422W71-112	15.250 (387.35)	14.000 (355.60)	10.000 (254.00)
NL422B71-167 NL422W71-167	22.500 (571.50)	20.875 (530.23)	15.000 (38.10)
NL422B71-207 NL422W71-207	27.500 (698.50)	25.875 (657.23)	20.000 (508.00)

1/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm. Metrics are in parentheses.

REQUIREMENTS:

Dimensions: See figure 1, tables I, and II.

Tolerances:
Ceramic - Unless otherwise specified, the tolerances for ceramic parts shall be as specified in table IV of MIL-I-23264.
Metal - Unless otherwise specified, the tolerance for metal parts shall be ±.020 inch. Tolerance on angles shall be ±1°.

Component parts:
Ceramic parts - Ceramic per MIL-I-10 shall be grade 422 or better.
Metal parts - Unless otherwise specified, all bearing and exposed metal surfaces shall be machine finished to 125/.

Tensile: The minimum working and ultimate tensile loads shall be as specified in table III, and shall be applied through end caps clamped into the tensile testing fixture.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T16:09Z
Check the source to verify that this is the current version before use.

(B) TABLE III. Tensile loads.

Type designation	Minimum working load (pounds) any insulator		Minimum ultimate load (pounds)	
	As received	After thermal shock	Any insulator	Average
NL422B68-208 NL422W68-208	3,000	---	---	----
NL422B69-136 NL422W69-136	2,000	---	---	---
NL422B70-104 NL422W70-104	2,000	---	---	---
NL422B70-168 NL422W70-168	2,000	---	---	---
NL422B71-072 NL422W71-072	2,000	---	---	4,000
NL422B71-112 NL422W71-112	2,000	---	---	4,000
NL422B71-167 NL422W71-167	---	3,500	4,000	7,000
NL422B71-207 NL422W71-207	---	3,500	4,000	7,000

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy - SH, EC
Air Force - 85
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5970-0517)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:09Z
Check the source to verify that this is the current version before use.